LESSON 7.4
ADDING INTEGERS USING NUMBER LINES

LESSON INTRODUCTION

MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.

LEARNING TARGETS

- I can add integers using a number line.

BENCHMARK COHERENCE

BUILDING ON

MA.5.NSO.2 Add, subtract, multiply, and divide multi-digit numbers.

WORKING TOWARD

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

ESSENTIAL QUESTIONS

- Where do negative integers fall on the number line? What about positive numbers?
- How can you represent adding integers using a number line?
- How do you use a number line to add integers?

LESSON NARRATIVE

In this lesson, students add integers using number line models. Students connect their prior work modeling integer addition using integer chips to represent integer addition on a number line. Students also review and apply the commutative property of addition.

COMPONENT SUMMARY

7.4.1 WARM-UP *(~5 MIN.)*

Watch a video about how speed is not important in math and reflect

7.4.2 REFRESHER *(5 MIN.)*

Plot positive and negative integers on a number line

7.4.3 GUIDED INSTRUCTION *(10 MIN.)*

Model integer addition with integer chips and on number lines

7.4.4 COLLABORATION *(10 MIN.)*

Apply the commutative property of addition to integer addition problems represented with integer chips, number lines, and equations

7.4.5 TRY IT *(5 MIN.)*

Practice adding integers using number lines

7.4.6 YOUR TURN! *(10 MIN.)*

Additional practice adding integers using number lines

7.4.7 WRAP-UP *(~5 MIN.)*

Self-assessment of understanding and confidence using number lines to add integers

RESOURCES

GLOSSARY

Key Terms:

- Commutative property of addition

Additional Terms:

- Integer
- Ray

MATERIALS

- (Optional) Integer chips
- (Optional) Colored pencils
- (Optional) Dry erase boards
- (Optional) Markers

ADDITIONAL PREPARATION

- 7.4.1 Warm-Up contains a video to accompany the question prompts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle to represent multiple addends on a number line, particularly where to begin the second addend.

THINGS TO CONSIDER

- Consider providing sentence stems for 7.4.1 Warm-Up and question 4 for 7.4.4 Collaboration.

LITERACY AND LANGUAGE ROUTINES

Take Turns

In 7.4.2 Refresher, students can first work independently to complete the problems. Then students can work with a partner and take turns sharing their responses as they review how to represent integers on a number line.

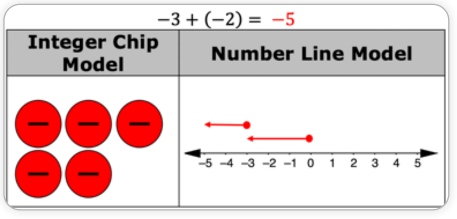
MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

In 7.4.3 Guided Instruction, students represent integer addition using integer chips, number lines, and numerical expressions. Ask students to share the similarities and differences among the representations and how they decide which representation(s) to use.

DIFFERENTIATE INSTRUCTION

Multiple representations (MA.K12.MTR.2.1) provide students the opportunity to demonstrate their understanding in a variety of ways, particularly finding models or modes that fit their understanding. Provide opportunities for students to engage with manipulatives, sketch and draft their own representations, and transition their representations from concrete to abstract. 7.4.3 Guided Instruction provides explicit models and samples for using representations to demonstrate understanding of adding integers. Encourage students not to isolate this approach to just this lesson but instead think creatively and strategically in which approach best fits their thinking and reasoning to apply throughout the unit.



7.4.1 WARM-UP: SPEED IS NOT IMPORTANT

Component Overview: *Play the video, "Speed is not Important" to launch the Warm-Up. Students learn that math involves deep thinking and therefore can be completed slowly. This component is designed to debunk the myth that speed is important in math.*

TEACHER MOVES

Think-Pair-Share

Have students share their responses with a partner after writing. Monitor for strong responses that highlight growth mindset and elicit those responses whole group. Also, prompt students to share something that their partner said that made them feel more confident or think about math in a different way.

7.4.1 Warm-Up: Speed Is Not Important

How fast you perform calculations in mathematics is not at all related to how "good" you are at math. Fast calculation is not important. What matters is deep thinking and understanding.

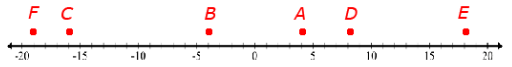
1. Describe a time when you felt rushed to perform calculations on a math assignment or assessment.
   Answers will vary.

2. If you were given as much time as you needed in the situation you described, how would that have affected your feelings and performance on completing the assignment or assessment?
   Answers will vary.

7.4.2 Refresher: Plotting Numbers on a Number Line

1. Plot each of the following integers on the number line below. Label each point with the letter indicated.

$A = 4 \quad B = -4 \quad C = -16 \quad D = 8 \quad E = 18 \quad F = -19$



2. Complete the statements.

   a. When moving from left to right (→) on the number line, the values
      o increase.
      o decrease.
      This can be described as moving in a
      o positive
      o negative
      direction.

   b. When moving from right to left (←) on the number line, the values
      o increase.
      o decrease.
      This can be described as moving in a
      o positive
      o negative
      direction.

7.4.2 REFRESHER: PLOTTING NUMBERS ON A NUMBER LINE

Component Overview: *Students activate prior knowledge of integers and plotting numbers on number lines by plotting positive and negative integers on a number line. Students fill in statements to review positive and negative directions on a number line.*

TEACHER MOVES

Take Turns

Students can work first independently to complete the problems. Then students can work with a partner and take turns sharing their responses as they review how to represent integers on a number line.

ENRICHMENT

Do not spend too much time on this Refresher as it is only meant to activate prior knowledge, not reteach a concept.

- What do you notice about the numbers to the left of the zero on the number line?
- What about the numbers to the right? What does this say about values as they go to the right on the number line?

7.4.3 GUIDED INSTRUCTION: RULES FOR ADDING INTEGERS ON A NUMBER LINE

Component Overview: *Students connect the integer-chip model to the number-line model, representing addition expressions using both. Students reason about how to accurately represent addends using rays and consider when number lines might be preferred representations.*

ENRICHMENT

MA.K12.MTR.2.1: Multiple Representations

Utilize inquiry to position as sense makers, connecting pictorial and concrete models and manipulatives to adding integers.

- How can you represent -3 on a number line? How can you represent adding a -2? Would you just put a dot on -2? Why or why not?
- How might you represent two values with different signs on a number line?
- Where is the sum located on the number line?

TEACHER MOVES

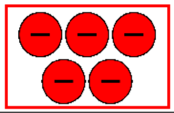
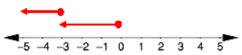
When providing Guided Instruction, make a note for students about how the blue and green paragraphs correspond with the blue and green arrows on the number line in question 3.

7.4.3 Guided Instruction: Rules for Adding Integers on a Number Line

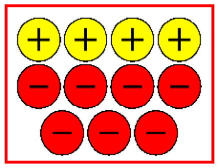
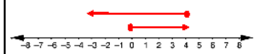
In addition to integer chips, number lines are another helpful tool you can use to model adding integers.

1. Complete the following to model integer addition.

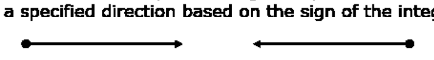
$-3 + (-2) = -5$

Integer Chip Model	Number Line Model
	

$4 + (-7) = -3$

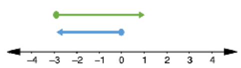
Integer Chip Model	Number Line Model
	

2. The sum of integers can be represented using rays on a number line. A ray has a single endpoint and extends in a specified direction based on the sign of the integer.



- A ray that represents a positive integer has its endpoint at 0, with the arrow extending to the *right*.
- A ray that represents a negative integer has its endpoint at 0, with the arrow extending to the *left*.
- To represent a sum, draw a ray for the first integer that starts at 0 and extends in the direction indicated by the *sign* of the first integer. It ends at the location of the first integer on the number line.
- Then, draw a ray for the second integer that starts where the first ray's arrow ends and extends in the direction indicated by the *sign* of the second integer. It ends after moving the number of units indicated by the second integer.
- The sum of the two integers is the integer where the second ray ends.

3. The diagram shown represents $-3 + 4 = +1$



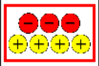
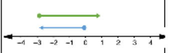
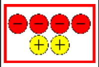
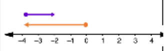
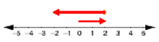
Give an example or explain when a number line model would be more useful than using integer chips.
Answers will vary. The number line model might be more useful when the integers are larger numbers, because there may be too many chips involved.

7.4.4 Collaboration: Commutative Property

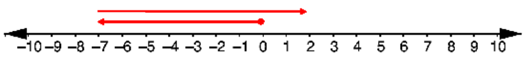
The ***commutative property of addition*** states that the order in which addition is performed does not change the value of the expression.

With your partner, add integers using the commutative property.

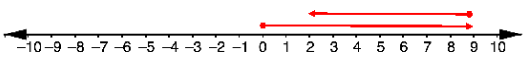
1. Complete the table by sketching the integer chip model for the first column, the number line model for the third column, and answering the question in the last column.

Integer Chip Model	Number Line Model	Commutative Property Number Line Model	Yes/No
$-3+4$ 	$-3+4$ 	$4+(-3)$ 	Is the sum of $-3+4$ the same as the sum of $4+(-3)$? *Yes*
$-4+2$ 	$-4+2$ 	$2+(-4)$ 	Is the sum of $-4+2$ the same as the sum of $2+(-4)$? *Yes*

2. Show the model for $(-7)+9$ on the number line below.



3. Show the model for $9+(-7)$ on the number line below.



4. Discuss with your partner how the two number line models for $(-7)+9$ and $9+(-7)$ show that $(-7)+9=9+(-7)$. Write a summary of your reasoning.

Answers will vary. In the number line model of $(-7)+9$, a ray begins at 0 and moves 7 units to the left, stopping at -7. The next ray beings at -7, and moving 9 units in the opposite direction, to the right, and stops at 2.

In the number line model of $9+(-7)$, a ray begins at 0 and moves 9 units to the right, stopping at 9. The next ray begins at 9, and moving 7 units, in the opposite direction, to the left, stops at 2.

In both models, the final location on the number line is 2.

7.4.4 COLLABORATION: COMMUTATIVE PROPERTY

Component Overview: *The commutative property of addition is defined and reviewed. Working with a partner, students practice modeling the commutative property of addition using number lines. Students discuss why the models of the commutative property are equivalent.*

ENRICHMENT

As students work on question 1, prompt them with the following:

- How can you represent 4 on the number line?
- Where should you start to represent -3?
- Which direction does a negative addend's arrow point?
- How do the number-line models show the sums of both addition expressions are the same?

ENRICHMENT

As students work on questions 2-4, prompt them with the following:

- Are (-7) + 9 and9 + (-7) equivalent? How do you know?
- How do the representations demonstrate this relationship? How does this show the commutative property of addition?

DIFFERENTIATE INSTRUCTION

Consider providing sentence stems on question 4 for students who struggle to begin writing their responses. Prompt students to use the words "equal to" or "equivalent" in context.

7.4.5 TRY IT: ADDING INTEGERS USING NUMBER LINES

Component Overview: *Students practice adding integers using number lines. Consider allowing students to work with a partner on this component.*

DIFFERENTIATE INSTRUCTION

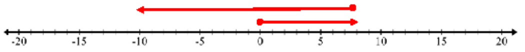
Consider providing tools and resources for students to successfully engage with this activity.

- Integer chips to verify their answers or represent their work in a different way
- Colored pencils to sketch integer chips or other creative models for integers

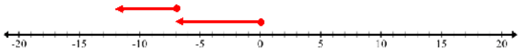
7.4.5 Try It: Adding Integers Using Number Lines

1. Find each sum. Use the number line to model the addition.

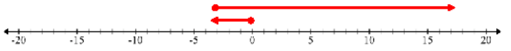
   a. $8 + (-18) = -10$



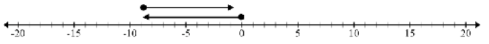
   b. $-7 + (-5) = -12$



   c. $-3 + 20 = 17$



2. Write the numerical expression for the model below. Then, find the sum.

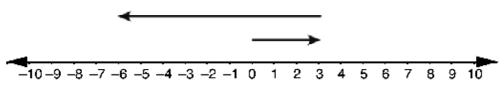


Numerical Expression	Sum
$-9 + 8$	-1

7.4.6 Your Turn!

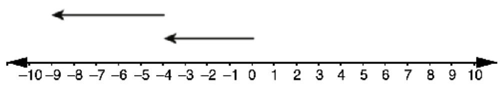
1. For each number line model, write the addition expression represented and find the sum.

a.



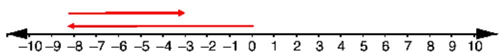
Addition Expression	Sum
$3 + (-9)$	-6

b.



Addition Expression	Sum
$(-4) + (-5)$	-9

2. Use the number line to show the sum of $^-8 + 5$. Then, complete the equation.



$^-8 + 5 = -3$

7.4.6 YOUR TURN!

Component Overview: *Students create addition expressions to match number line models and find sums. Students represent addition of temperatures on a vertical number line in context.*

DIFFERENTIATE INSTRUCTION

Consider providing tools and resources for students to successfully engage with this activity.

- Use integer chips to verify their answers or represent their work in a different way.
- Use colored pencils to sketch integer chips or other creative models for integers.

7.4.6 YOUR TURN! (CONTINUED)

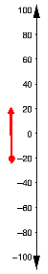
DIFFERENTIATE INSTRUCTION

This is the only vertical number line depicted in this lesson. Consider providing real-world examples of vertical number lines, such as a thermometer, to activate students' prior knowledge and provide an entry point.

QUESTIONING

To add complexity, consider having students who finish early write additional real-world examples of integer addition. Then ask them to represent the addition on a number line and with an equation.

3. In a single day, the temperature in degrees Fahrenheit (°F) of a town in Canada began at a low of -20°F and increased 40°F by noon.



   a. Plot a point at the low temperature.

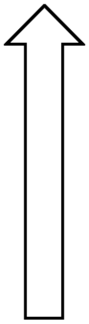
   b. Draw a ray along the number line to represent the increase of 40°F.

   c. Which direction does the arrow extend to show the increase in temperature?
   Up

   d. What was the temperature in the town by noon?
   20°F

7.4.7 Wrap-Up: Reflection

1. Color in the arrow up to the statement that best describes your current level of confidence with using a number line to add integers.
   Answers will vary.



I'm very confident! I can teach anyone how to add integers.

I can help others add integers.

I can add integers on my own.

I can add integers with assistance.

I'm having trouble adding integers.

I'm not sure what integers are.

7.4.7 WRAP-UP: REFLECTION

Component Overview: *Students self-assess their understanding and confidence with using number lines to add integers.*

DIFFERENTIATE INSTRUCTION

For students who color the arrow from "I can add integers with assistance" and below, consider prompting those students to provide an example of where they have trouble or what questions they still have.

For students who color the arrow from "I can add integers on my own" and above, consider prompting them to provide an example or write an explanation to a peer or younger family member.